Lukas Klein

+49 157 89652315 lukas.klein96@outlook.de lukaskln.github.io

Work Experience

German Cancer Research Center (DKFZ) *08/2021 - Today*
Graduate Researcher - Supervised by Dr. Jäger (Google DeepMind) and Prof. Maier-Hein (DKFZ).
- Machine Learning Lead in several interdisciplinary research projects (Biomedical, Material Science, Natural Language).
- Responsible for model training within multi-GPU HPC environments, as well as data preprocessing, MLOps, and model evaluation.
- Developing computer vision (video, (fluorescence-)microscopy, MRI, CT, (multispectral-)imaging), sequence (protein, RNA, timeseries, NLP), spatial transcriptomics and vision-language (generative) models in Python.

Eidgenössische Technische Hochschule (ETH) Zürich *02/2021 - 08/2021*
Research Intern - Institute for Machine Learning supervised by Prof. Buhmann.

Camelot ITLab *01/2019 - 08/2019*
IT Consultant - Machine Learning and cloud solutions in the pharma and chemical industry.

Camelot ITLab *01/2018 - 01/2019*
Thesis and Working Student - Bachelor Thesis and Data Science consulting.

Microsoft *06/2017 - 09/2017*
Intern - National Technology Office supervised by Thomas Langkabel.

Education

Eidgenössische Technische Hochschule (ETH) Zürich *2021 - Today*
Ph.D. in Computer Science – Supervised by Prof. Buhmann, Prof. Krause and Dr. Jäger.
- External group member in the IVIA Lab of Prof. El-Assady.
- Collaboration with the MIT-IBM Watson AI Lab supervised by Dr. Strobelt.

Helmholtz International Graduate School for Cancer Research *2021 - Today*
DKFZ International Scholarship Award - Acceptance rate of 2.7% in 2022.
- Completed the DKFZ International PhD Program for Cancer Research.

University of Oxford *2022*
OxML Summer School for Machine Learning in Healthcare.

University of Geneva *2019 - 2021*
M.Sc. in Statistics – Final grade: Very Good. Master Thesis and Exchange Semesters at ETH Zürich.

University of Mannheim *2015 - 2018*
B.Sc. in Economics - Final grade: Good. Exchange Semesters at the Norwegian School of Economics.

Languages & Skills

Languages: German (Native), English (Fluent), French (Basic)

Programming Languages:	
Python	Expert
R	Expert
Bash	Expert
Git	Expert
LaTeX	Advanced
SQL	Advanced
CUDA	Basic
Julia	Basic

Libraries:	
PyTorch	Expert
Hydra	Expert
Numpy	Expert
XGBoost	Expert
SciKit	Expert
Transformers	Expert
Torchvision	Expert
SciPy	Advanced
Lme4 (R)	Advanced
Tidyverse (R)	Advanced
JAX	Basic
TensorFlow	Basic

Software & Platforms:	
Linux (Ubuntu, CentOS, Debian)	Expert
GitHub & GitLab (with CI & CD)	Expert
IBM Spectrum LSF	Expert
QuPath / ImageJ	Expert
Weights & Biases	Expert
Docker/Apptainer Container	Advanced
Nvidia Jetson IoT Platform	Advanced
Azure (ML Studio, OpenAI Services)	Advanced
Google Cloud (Cloud Build & Run, Vertex AI)	Advanced
SAP Analytics Cloud	Advanced
CMake	Basic
NCI Genomic Data Commons & TCGA	Basic

Voluntary

2022 – Today **Advisory Committee of the Oxford ML Summer School**
AI for Global Goals.

2022 – Today **Invited Talks and Seminars**
TU Munich, DESY, HIDA Graduate School, HIDSS4Health Graduate School, IBM Zürich, MIT-IBM Watson AI Lab, Harvard University, MIT, Tufts University

2022 – 2023 **Teaching**
Advanced Topics in Machine Learning (ETH Zürich).

2022 – 2023 **Conference Reviewer**
ICML, ICLR, EuroVis, NeurIPS Workshop on RedTeaming GenAI

2016 – 2017 **Speaker for Public Relations**
AStA (Student Council) at the University of Mannheim.

2015 – 2016 **Member of the Student Parliament and Deputy Member of the Information Committee**
University of Mannheim. Representative for Refugee Projects.

Publications

Google Scholar: https://scholar.google.com/citations?user=sxpuZq4AAAAJ

2024 **Navigating the Maze of Explainable AI: A Systematic Approach to Evaluating Methods and Metrics**
Lukas Klein, Carsten Lüth, Udo Schlegel, Till Bungert, Mennatallah El-Assady, Paul Jäger. *Venue: NeurIPS 2024.*

2024 **Explainable AI-based analysis of human pancreas sections identifies traits of type 2 diabetes**
Lukas Klein, Sebastian Ziegler, Yanni Morgenroth, Eyke Schoeniger, Michele Solimena, Felicia Gerst, Robert Wagner, Fabian Isensee, Klaus Maier-Hein, Paul Jäger. *Venue: Under review at Nature Metabolism.*

2024 **Why context matters in VQA & Reasoning: Semantic interventions for VLM input modalities**
Lukas Klein, Kenza Amara, Carsten Lüth, Antonio Rodríguez, Hendrik Strobelt, Mennatallah El-Assady, Paul Jäger.
Venue: Under review at ICLR 2025.

2024 **Enhancing predictive imaging biomarker discovery through treatment effect analysis**
Shuhan Xiao, **Lukas Klein**, Paul Jäger, Jens Petersen, Klaus Maier-Hein. *Venue: WCAV 2025.*

2024 **Interactive Semantic Interventions for VLMs: Breaking VLMs with Human Ingenuity**
Lukas Klein, Kenza Amara, Carsten Lüth, Antonio Rodríguez, Hendrik Strobelt, Mennatallah El-Assady, Paul Jäger.
Venue: NeurIPS 2024 Safe Generative AI Workshop and *Causality & Large Models Workshop.*

2024 **Explainable AI-based analysis of human pancreas sections detects traits of type 2 diabetes**
Lukas Klein, Sebastian Ziegler, Yanni Morgenroth, Eyke Schoeniger, Michele Solimena, Felicia Gerst, Robert Wagner, Fabian Isensee, Klaus Maier-Hein, Paul Jäger. **Venue:** *NeurIPS 2024 IAI Workshop.*

2023 **Discovering Process Dynamic for Scalable Perovskite Solar Cell Manufacturing with Explainable AI**
Lukas Klein, Sebastian Ziegler, Felix Laufer, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich W. Paetzold, Fabian Isensee, Paul Jäger. *Journal: Advanced Materials (IF: 30).* ***This work has been covered i.a. by The Independent.***

2023 **Understanding Solar Cell Manufacturing with Explainable AI**
Lukas Klein, Sebastian Ziegler, Felix Laufer, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich W. Paetzold, Fabian Isensee, Paul Jäger, *Venue: NeurIPS 2023 Workshop on XAI in Action* ***(Oral, Top 3 Paper).***

2023 **Navigating the Pitfalls of Active Learning Evaluation: A Systematic Framework for Assessment**
Carsten T. Lüth, Till J. Bungert, **Lukas Klein**, Paul Jäger. *Venue: NeurIPS 2023.*

2023 **Semi-supervised automated Gleason Grading on WSI**
Maximilian Fischer, Lars Krämer, Sebastian Ziegler, **Lukas Klein**, Paul Jäger, Marco Nolden, Peter Neher, Fabian Isensee, Klaus Maier-Hein, *Venue: ECDP 2024 (Oral).*

2023 **A Call to Reflect on Evaluation Practices for Failure Detection in Image Classification**
Paul Jäger, Carsten T. Lüth, **Lukas Klein**, Till J. Bungert. *Venue: ICLR 2023* ***(Oral, Top 1% Paper).***

2022 **From Correlation to Causation: Formalizing Interpretable Machine Learning as a Statistical Process**
Lukas Klein, Mennatallah El-Assady, Paul Jäger. *Venue: IJCAI 2022 Workshop on XAI.*

2022 **Improving Explainability of Disentangled Representations using Multipath-Attribution Mappings**
Lukas Klein, João Carvalho, Mennatallah El-Assady, Paolo Penna, Joachim M. Buhmann, Paul Jäger. *Venue: MIDL 2022.*

References

Prof. em. Joachim Buhmann
Professor at ETH Zürich leading the ISE Group.
jbuhmann@inf.ethz.ch

Dr. Paul Jäger
Led the IML Group at the DKFZ. Google DeepMind Senior Researcher.
p.jaeger@dkfz-heidelberg.de

Prof. Mennatallah El-Assady
Professor at ETH Zürich leading the IVIA Lab.
menna.elassady@ai.ethz.ch